

Study Questions: Reasoning About Results

Analytical Questions

1. Describe the Bayesian updating process for an inventor evaluating test results. What are the prior, likelihood, and posterior in this context?
2. What does "proportional updating" mean, and why do inventors commonly violate it in both directions (ignoring evidence and overreacting to evidence)?
3. Explain the difference between signal and noise in prototype test data. What practical indicators help an inventor distinguish between them?
4. How does Deming's distinction between "common cause variation" and "special cause variation" apply to prototype testing? What happens when an inventor treats noise as signal?
5. Define parameter updating, structure revision, and model replacement. Give an invention example of each.
6. Explain confirmation bias as a Bayesian phenomenon. How does an asymmetric likelihood function lead to biased reasoning about test data?
7. What is motivated reasoning, and how does it corrupt the balance between epistemic and pragmatic value in Active Inference?
8. Describe the Analysis of Competing Hypotheses method. Why is it more rigorous than evaluating a single preferred hypothesis?
9. What is pre-registration and why has it been mandated in clinical trials? How can inventors apply the same principle?
10. How does anchoring bias affect the interpretation of sequential test results? What is the Bayesian explanation for why early data has disproportionate influence?

Applied Questions

1. You test 10 prototypes of a new battery design. Seven last 8+ hours (your target) and three last only 5 hours. Your prior belief was that all would last 8+ hours. Walk through the Bayesian update: what should your posterior belief be? Is the 30% failure rate signal or noise?
2. Your co-founder presents test results showing that your prototype app has a 4.2/5 user satisfaction score. You are elated. Apply the bias audit framework: what cognitive biases might be inflating your interpretation of this result?
3. An inventor tests a waterproof phone case by submerging it in water for 30 minutes. One out of five cases leaks slightly. The inventor concludes the design is sound because 80% passed. Evaluate this reasoning using Bayesian principles and the concept of base rate neglect.
4. You receive contradictory feedback: engineers love your prototype's mechanism but users say it is too complicated. Using the concept of competing hypotheses, list at least three possible explanations and specify what additional evidence would discriminate between them.
5. A pharmaceutical company tests a drug on 1,000 patients. The drug produces a statistically significant improvement of 2% over placebo. Should they proceed to market? Analyze this using the concepts of signal precision, practical significance vs. statistical significance, and proportional updating.
6. You have been working on a robotic arm prototype for two years. The latest test shows that the arm fails to meet accuracy requirements by a significant margin. Analyze the sunk cost bias that might affect your reasoning about whether to continue, pivot, or abandon the project.
7. Design a pre-registration document for testing a new type of bicycle lock. Include: the hypothesis, specific test procedures, success criteria, failure criteria, and a decision matrix mapping results to actions.

8. Two investors review your test data. One sees a promising trend; the other sees inconclusive noise. Using the concept of different generative models and precision weighting, explain how both interpretations can be rational given different priors.
9. The Challenger disaster involved a failure to reason correctly about O-ring data under pressure. Identify the specific reasoning errors (base rate neglect, burden of proof inversion, motivated reasoning) and propose a structured reasoning protocol that could have prevented the decision to launch.
10. You notice that your prototype performs better in the morning than in the afternoon during testing. Generate three competing hypotheses for this pattern, evaluate each against the evidence, and design a test to discriminate between them.